Proof by Induction: Matrices

Questions 1–15

Question 1. Show that

a) $\begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 2k+1 & -4k \\ k & 1-2k \end{pmatrix} = \begin{pmatrix} 2(k+1)+1 & -4(k+1) \\ k+1 & 1-2(k+1) \end{pmatrix}$

b) $\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 1 & 2^k-1 \\ 0 & 2^k \end{pmatrix} = \begin{pmatrix} 1 & 2^{k+1}-1 \\ 0 & 2^{k+1} \end{pmatrix}$

c) $\begin{pmatrix} 1 & 0 \\ -1 & 5 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ -\frac{1}{4}(5^k-1) & 5^k \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -\frac{1}{4}(5^{k+1}-1) & 5^{k+1} \end{pmatrix}$

Question 2. Prove by induction, for all integers $n \geq 1$, that $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$.

Question 3. Given $A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$, prove for all integers $n \geq 1$ that $A^n = \begin{pmatrix} 1 & 2n \\ 0 & 1 \end{pmatrix}$.

Question 4. Prove by mathematical induction, for $n \in \mathbb{N}$, that $\begin{pmatrix} 3 & -2 \\ 2 & -1 \end{pmatrix}^n = \begin{pmatrix} 2n+1 & -2n \\ 2n & 1-2n \end{pmatrix}$.

Question 5. Prove by mathematical induction, for $n \in \mathbb{N}$, that $\begin{pmatrix} 3 & 2i \\ 2i & -1 \end{pmatrix}^n = \begin{pmatrix} 2n+1 & 2ni \\ 2ni & 1-2n \end{pmatrix}$.

Question 6. Prove by mathematical induction, for $n \in \mathbb{N}$, that $\begin{pmatrix} 2 & 2 \\ 0 & 4 \end{pmatrix}^n = \begin{pmatrix} 2^n & 4^n - 2^n \\ 0 & 4^n \end{pmatrix}$.

Question 7. Given that $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$, prove for $n \in \mathbb{N}$ that $A^n = \begin{pmatrix} 2^n & n2^{n-1} \\ 0 & 2^n \end{pmatrix}$.

Question 8. Given that $A = \begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix}$, prove for $n \in \mathbb{N}$ that $A^n = \begin{pmatrix} 1 & \frac{1}{2}(3^n - 1) \\ 0 & 3^n \end{pmatrix}$.

Question 9. Prove by mathematical induction, for $n \in \mathbb{N}$, that $\begin{pmatrix} 1 & 0 \\ -1 & 4 \end{pmatrix}^n = \frac{1}{3}\begin{pmatrix} 1 & 0 \\ 1-4^n & 4^n \end{pmatrix}$.

Question 10. Given that $A = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$,

a) Find A^2, A^3 and A^4.
b) Write down a suitable form for A^n where n is a positive integer.
c) Prove your answer to part b).

Question 11. Given that $A = \begin{pmatrix} 4 & a \\ 0 & 1 \end{pmatrix}$ where a is real, prove $A^n = \begin{pmatrix} 4^n & \frac{a}{3}(4^n - 1) \\ 0 & 1 \end{pmatrix}$ is true for all positive integers n.

Question 12. Given that $A = \begin{pmatrix} -2 & -9 \\ 1 & 4 \end{pmatrix}$,

a) Prove that for $n \in \mathbb{N}$, $A^n = \begin{pmatrix} 1 - 3n & -9n \\ n & 3n + 1 \end{pmatrix}$.

b) Find the line of invariant points of the transformation represented by matrix A.

c) Show, algebraically, that the transformation A^n has the same line of invariant points.

Question 13. It is known that $A = \begin{pmatrix} 2 & -a \\ a & 2 \end{pmatrix}$ where a is imaginary and $\det A = 0$.

a) Find the value of a.

b) Prove by mathematical induction, for $n \in \mathbb{N}$, $A^n = 4^{n-1}A$.

Question 14. Two square matrices, A and B, are such that $AB = B^4A$ and $B^5 = I$. Prove that $B^nAB^n = A$ for $n \in \mathbb{N}$.

Question 15. Prove by mathematical induction, for $n \in \mathbb{N}$, that

a) $\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}^n = \begin{pmatrix} 1 & n & \frac{1}{2}n(n+1) \\ 0 & 1 & n \\ 0 & 0 & 1 \end{pmatrix}$

b) $\begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}^n = \begin{pmatrix} 1 & 2n & \frac{1}{2}n(n+3) \\ 0 & 1 & n \\ 0 & 0 & 1 \end{pmatrix}$

c) $\begin{pmatrix} 2 & -1 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}^n = \begin{pmatrix} 2^n & 1-2^n & 3(2^n-1) \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$